

NeuroShift 2.0 - Complete Guide Brief

Single document covering base paper extension, novelty, phases, hardware, Indian home integration, cost, PwD scenarios, cameras, and population need | July 2026

Principle: Looking selects. Muscle confirms. Unsure -> Abstain. Novelty: gaze-selected home intention + EMG confirmation + Act/Abstain + FAR metric.

0. 5-minute pitch for guide

- Problem: motor-impaired users struggle with switches; wrong actuations are unsafe.
- Base paper (EMBC 2024): EMG + vision improves prosthesis GESTURE recognition.
- Our idea: camera detects appliances + gaze selects target; EMG confirms toggle; Act/Abstain.
- Sem 7: working demo (MyoWare + webcam + ESP32 relay hub, 2-3 devices).
- Sem 8: eye-gaze upgrade, evaluation, IEEE-style paper.
- Ask: approve scope, ethics, Sem 7 deliverables, title freeze.

1. Why this project is needed (population data)

Metric	Number	Source type
People living with stroke effects	~94-105 million	WHO / GBD / WSO
New strokes per year	~12-13 million	WHO / GBD
Lifetime stroke risk	1 in 4 adults (25+)	WSO / WHO
Stroke economic cost	> US\$890B / year	WSO / Lancet
Spinal cord injury survivors	~20.6 million (2019)	GBD
Traumatic limb amputees	~57.7 million (2017)	GBD-based study
India persons with disability	26.8 million	Census 2011
Movement disability share (India)	~20.3% of PwD	Census 2011
India stroke prevalence	~9.4 million (2021)	GBD / Lancet reporting
India new strokes (2021)	~1.25 million (~10% global)	GBD / Lancet reporting

NeuroShift targets independent living assistance for people with residual voluntary muscle activity who cannot easily use remotes/switches.

2. Base paper + our extension

Base paper

Visual Scene Understanding for Enhanced EMG Gesture Recognition | Chamberland, Labbe, Tam, Scheme, Gosselin | IEEE EMBC 2024 | DOI 10.1109/EMBC53108.2024.10782354

- EMG (EMaGer 64-ch HD-EMG) + head webcam + YOLOv8 + MediaPipe Hands
- SDCNN classifies 6 hand grips; vision filters object-compatible grips (OLF)
- Domain: myoelectric prosthesis control
- Result: fewer false gesture detections in pilot trials

What they do NOT solve

- Smart-home appliance control
- Gaze-based appliance selection
- Act/Abstain safety for IoT
- False Actuation Rate for home devices
- Low-cost sparse EMG + Indian wired homes

Our extension (one paragraph)

Chamberland et al. use vision to rescue EMG gestures for prosthesis grips. NeuroShift extends this multimodal idea to assistive smart homes: the camera detects appliances and estimates which one the user is looking at; sparse EMG confirms whether they truly want to toggle it; the system Acts or Abstains to minimize False Actuation Rate.

Point	Base paper	NeuroShift
Predicts	Gesture / grip class	Home device action or Abstain
Vision role	Validate grips	Detect appliances + gaze-select target
EMG role	Classify gesture	Confirm volition only
Selector	Hand near object	Head/eye gaze (primary)
Safety	Fewer false movements	Two-factor + Act/Abstain + FAR
Hardware	Custom 64-ch HD-EMG	MyoWare + 2xESP32 + relays
Setting	Prosthesis lab scenarios	Indian wired-home appliances

3. Novelty (locked)

Looking selects. Muscle confirms. Unsure -> Abstain.

Novelty piece	Meaning
Gaze-selected home intention	Camera finds appliances AND which one user is looking at
EMG confirmation	Muscle effort confirms real intent (not a gesture dictionary)
Act / Abstain	If gaze or EMG unclear, do nothing (safer than guessing)
FAR metric	Primary success = False Actuation Rate, then intention F1

We do NOT claim: first EMG+camera fusion, Ninapro SOTA, or Transformer novelty alone.

Official real-time pipeline

Webcam -> YOLO appliances + Gaze/head direction
-> Match gaze to appliance (dwell 0.5-1.0 s) -> SELECTED TARGET
MyoWare EMG -> Volition confirm
If gaze stable AND EMG high -> ACT (MQTT -> ESP32 -> relay)
Else -> ABSTAIN
Dashboard logs: target, EMG, decision, reason

4. Phase 1 (Sem 7) and Phase 2 (Sem 8)

	Phase 1 / Sem 7	Phase 2 / Sem 8
Goal	BUILD working prototype + live demo	PROVE research + IEEE draft
Gaze	Head/face direction (MediaPipe)	Eye-gaze model + dwell calibration
Fusion	Rule: gaze AND EMG -> Act else Abstain	Learned fusion + FAR-tuned abstention
Extra	Logger + simple dashboard	Habit prior + decision XAI + caregiver UI
Demo	2-3 appliances, Act + Abstain scenes	Full eval tables + polished demo
Success	External sees reliable live control	Lower FAR vs baselines + paper

Phase 1 algorithms

- Objects: YOLO11 / YOLOv8 / RT-DETR
- Gaze: MediaPipe Face Mesh / head pose
- EMG: RMS/MAV threshold or small TCN (confirm only)
- IoT: MQTT + ESP32 relay hub

Phase 2 algorithms

- Eye-gaze estimation; calibrated volition; small fusion network
- Personal temporal prior; decision-level explanations
- Baselines: EMG-only, gaze/vision-only, Chamberland-style gate, always-act

Out of scope (say to guide)

- Wheelchair control, hospital system, fall/medicine as Paper-1 core
- Custom HD-EMG like EMaGer; Tobii eye tracker; cloud LLM as sole brain

5. Hardware stack

[MyoWare on arm] --analog--> [ESP32 #1 battery] --Wi-Fi EMG--> [Laptop AI]
Webcam (USB near user) -----> [Laptop AI]
[Laptop AI: YOLO + Gaze + EMG + Act/Abstain + Dashboard]
--Wi-Fi MQTT--> [ESP32 #2 + 8/16-ch relay hub] --> Lamp/Fan/Plug

Item	Role	Notes
MyoWare 2.0 + electrodes	EMG volition	Not USB-direct; needs ESP32 ADC
ESP32 #1	Wearable EMG node	Battery/power bank
ESP32 #2	Relay hub	5V adapter; Wi-Fi MQTT
8/16-ch relay board	Switch appliances	One ESP32 controls many devices
1080p USB webcam	Face + zone appliances	C920-class preferred; short USB
Laptop	Main NeuroShift app	Camera, AI, MQTT, dashboard
Lamp + fan (demo)	Loads	Via relay demo hub / smart plug

USB is for firmware/debug and short webcam cable. Final demo communication is Wi-Fi. Do not use one ESP32 per appliance.

6. Indian wired-home integration

Most Indian homes have normal switches, not smart bulbs. NeuroShift adds a relay/smart-switch layer on existing wiring.

Option A (Phase 1 demo) - portable relay hub

- Wall socket -> Demo Hub (ESP32 + relays) -> lamp/fan plugs
- No wall switchboard opening required
- Wireless decisions over MQTT; hard switching inside enclosed box

Scaling to many appliances

- 1 ESP32 + 8/16-channel relay (cheap) OR modular MQTT smart switches
- AI publishes topic per device; channels scale without redesigning AI

Safety

- Optocoupler relays, fuse, enclosure, manual override, electrician for mains

7. Cost and feasibility

Budget style	Approx total (INR)
Lean Phase 1	7,000 - 9,000
Recommended Phase 1	10,000 - 15,000
Comfortable + camera upgrade	15,000 - 22,000
Software stack	0 (open source)

Goal	Feasible?
Live Act/Abstain demo Sem 7	Yes - High
IEEE-style Sem 8 evaluation + draft	Yes - Medium-High
Clinical medical product	No - out of scope now

8. PwD scenarios and limits

Condition	How NeuroShift works	Fit
Hand weakness / tremor	Forearm EMG + gaze select	Excellent
Stroke hemiparesis	EMG on stronger side + gaze	Very good
Arm amputation	Residual stump EMG + gaze	Excellent
Partial upper paralysis	Shoulder/upper EMG if available	Good if signal exists
Paraplegia (arms OK)	Reach assistance for switches	Good
Severe quadriplegia	Only if residual neck/shoulder EMG exists	Conditional
No voluntary EMG	Needs gaze/voice/other modalities later	Not Phase 1 claim

Correct claim: for motor impairment with some voluntary EMG. Incorrect claim: works for all paralysis types automatically.

9. Camera FOV, count, and multi-zone

A ~78 degree webcam cannot cover an entire 8-16 m room alone. Use zones.

Buy recommendation

- Phase 1: 1x 1080p autofocus USB webcam (Logitech C920/C922 class) near laptop
- Phase 2: optional 2nd room/IP/phone camera for far appliances
- Do not buy Tobii / ESP32-CAM for gaze core

USB length

- Keep face cam on short USB (1-2 m). Far views use Wi-Fi phone/IP cams.

Multi-zone home

Face cam (near user) -> gaze
 Room cam A/B/C (Wi-Fi zones) -> appliances
 Laptop merges detections; gaze selects; EMG confirms; Act/Abstain

10. External demo scenes (must rehearse)

Scene	What happens
A Clear Act	Gaze on lamp + EMG effort -> lamp ON
B Abstain look-only	Gaze on fan, no EMG -> ABSTAIN
C Abstain EMG-only	Muscle noise, no stable gaze -> ABSTAIN
D Ambiguity	Two devices near gaze -> ABSTAIN

11. Ask guide to confirm

- Ethics/consent for pilot subjects (team vs real PwD)
- Exact Sem 7 deliverables (demo, report, PPT, lit survey length)
- Team role split
- Lab already has webcam/ESP32/relays?
- Approve title and out-of-scope list
- Electrical safety: electrician for mains relay box?

Proposed title

NeuroShift: Gaze-Selected and EMG-Confirmed Smart-Home Intention Inference with Selective Actuation for Assistive Living

12. Bottom line

- Base paper = safer prosthesis gestures with EMG+vision.
- NeuroShift = safer home control with gaze-select + EMG-confirm + Act/Abstain.
- Primary metric = False Actuation Rate (FAR).
- Sem 7 builds demo; Sem 8 proves and writes paper.
- Low-cost MyoWare + 2xESP32 + webcam + relay hub is feasible (~Rs 10-15k).

Ready for guide discussion on concept. Remaining items are logistics: ethics, Sem 7 marking deliverables, shopping, and safety sign-off.